

Aim

To observe and study biofertilizers, namely *Azotobacter*, *Nostoc*, *Anabaena* and *Azolla*.

Requirements

Fresh specimens of *Azolla* along with permanent slides of *Azotobacter*, *Nostoc* and *Anabaena*.

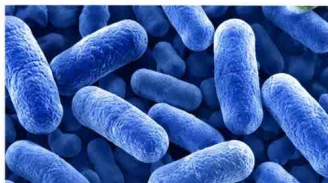
Procedure

Carefully examine the *Azolla* specimens and the permanent slides of the remaining three organisms.

(Note: Record your observations along with the agricultural uses of these organisms.)

Observation Table

1. *Azotobacter*



1. *Azotobacter*

Kingdom: Monera | Division: Proteobacteria

Underline the correct word:

Oval and round/~~rectangular~~ in shape.

Develop hard cyst/~~crust~~ around.

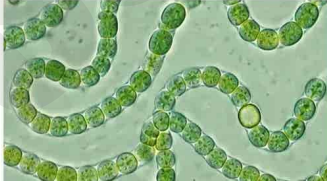
Freely living/~~parasitic~~ in soil.

Uses:

1. Ability to fix molecular nitrogen/~~nitrogen compounds~~ and therefore increase the soil fertility and stimulate overall growth/~~only height~~ of plant.

2. *Azotobacter* species are widely used in agriculture, particularly in nitrogen-fixing biofertilizers such as azotobacterin/~~antibiotic bacterin~~.

2. *Nostoc*



2. *Nostoc*

Kingdom: Monera | Division: Cyanobacteria

Underline the correct word:

Found in ~~water~~/various environments that forms colonies.

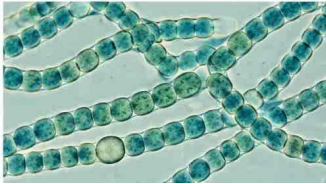
Found in soil, on moist/~~dry~~ rocks, at the bottom of lakes and springs, both fresh and saltwater. It is free living/~~parasitic~~.

Also grows ~~parasitically~~/symbiotically in the plants' tissues.

Uses:

1. *Nostoc* contains two pigments/~~proteins~~, blue/~~black~~ phycocyanin and red/~~yellow~~ phycoerythrin, as well as green/~~brown~~ chlorophyll.
2. It can fix nitrogen in specialized cells called heterocysts/~~rhizoids~~.
3. Also used as ~~tasty~~/supplementary food due to vitamins and proteins/~~carbohydrates~~ in them.

3. *Anabaena*



3. *Anabaena*

Kingdom: Monera | Division: Cyanobacteria

Underline the correct word:

Filamentous cyanobacteria that exist as

phytoplankton/~~zooplankton~~.

Can live symbiotically with Azolla/~~mushroom~~.

Uses:

1. Found in paddy as natural/~~chemical~~ fertilizer.
2. They produce toxic neurotoxins/~~haemotoxins~~ and hence are useful to keep away the grazing animals/~~birds~~ from farms.

4. Azolla



4. Azolla

Kingdom: Plantae | Division: Pteridophyta

Underline the correct word:

Azolla ~~sinks in deep water~~/floats on the surface of water

by means of numerous, small, closely overlapping scale-like leaves.

Their roots are hanging in the water/~~air~~.

Uses:

1. They can fix atmospheric nitrogen/~~carbon dioxide~~.
2. Traditional/~~Modern~~ cultivation as a biofertilizer for paddy.
3. Used as livestock/~~infant~~ feed.
4. Azolla is rich in proteins, essential amino acids/~~carbohydrates~~, vitamins and minerals.

Inference

Biofertilizers are a non-polluting option for increasing crop production.

Multiple Choice Questions

Choose the correct alternative and write its letter (A), (B), (C), (D) in the box:

1. Which of the following has the wormicidal properties?

(A) Azolla (B) Anabaena (C) Nostoc (D) Azotobacter

Answer: (A) Azolla

2. Which of the two show symbiotic relationship with each other?

(A) Nostoc and Anabaena (B) Nostoc and Azotobacter
(C) Azolla and Anabaena (D) Azolla and Nostoc

Answer: (C) Azolla and Anabaena

3. Find the odd one out from the viewpoint of classification.

(A) Azolla (B) Anabaena (C) Nostoc (D) Azotobacter

Answer: (A) Azolla

4. Which of the following is autotrophic bacteria?

(A) Azolla (B) Anabaena (C) Nostoc (D) Azotobacter

Answer: (A) Azolla

5. Why these organisms are said to be important in agriculture?

- (A) They do not cause pollution.
- (B) They increase the yield of the crops.
- (C) They are natural and biofertilizers.
- (D) All of the above.

Answer: (D) All of the above.

Remember

Biofertilizers are microbes which promote plant growth by increasing nutrient availability in soil.

Date: _____

Teacher's Signature: _____